

Aligning two traversals of the same route

Measured, then frozen, then tested once — including the tests that failed

Outcome. Frame identity is the zero-based display-order image from one canonical decoder, verified against the bitstream rather than assumed, and each camera is aligned independently. Against 40 prediction-blind labels per camera, the submitted mapping places 84.4% of accepted CAM0 matches and 68.8% of CAM5 matches within ± 2 frames of the annotator's interval (0.2 s, about 2 m) at 100.0% and 86.5% coverage, and 96.9%/96.9% within ± 10 frames, the field's usual tolerance; the sparse-anchor method it replaces answered only 53.1% and 10.8% of the queries. A second, targeted blind set reads 81.0% (CAM0) and 71.4% (CAM5, re-annotated after a tool defect) at the same tolerance and locates most of CAM5's residual in a two-frame difference between two definitions of "same place". The strict single-frame reading (0.719/0.375) is the last table.

1 Characterising the recordings

Provenance. Picture counts, rasters and picture order are parsed from the HEVC Annex-B bitstream by a purpose-written scanner, independently of libavcodec; the two agree exactly on all four files. Nominal 10 FPS is stream-declared; timestamp statistics come from the integer-nanosecond sidecars and describe capture-log events, not decoded-video cadence.

Run	Camera	Pictures	File size (B)	Coded	Display	Reordered
A	CAM0	2,674	128,974,848	1440×1088	1440×1080	74.4%
A	CAM5	2,674	121,896,960	1440×1088	1440×1080	74.4%
B	CAM0	2,622	69,992,448	1440×1088	1440×1080	74.4%
B	CAM5	2,615	63,963,136	1440×1088	1440×1080	74.4%

Run A: 2025-02-28 06:24:09.580 UTC, span 269.595 s, 2,695 sidecar rows; Run B: 2024-04-13 09:12:41.985 UTC, 266.492 s, 2,663 rows — eleven months *earlier*, so the run names carry no chronology. Cadence 9.9928 and 9.9890 Hz measured, both logs strictly increasing. A road sign (*Kawasan SD 13*, Bandar Sri Damansara) puts the route in Malaysia: UTC+8, local 14:24 and 17:12, inferred from signage, not declared.

1.1 Four findings that silently corrupt data

- **Decode order is not display order for 74.4% of pictures.** A four-frame B-pyramid: bitstream order runs 0, 4, 2, 1, 3, 8, 6, 5, 7. Any tool calling the n -th packet "frame n " scrambles the sequence locally, by the same magnitude as the error being measured. The pipeline reconstructs picture order and *fails* unless display order is a clean permutation.
- **The declared frame rate contradicts itself.** FFprobe reports `r_frame_rate` 10/1 from the stream and `avg_frame_rate` 25/1 as its own fallback. Both are declared; neither is measured.
- **Colour is undeclared.** Only `color_range=tv` is signalled; `swscale` picks a matrix from frame height (BT.709 for the source, BT.601 for a 480-tall proxy). Pinning it changes 3.17% of decoded bytes; cache keys carry source hash, raster and colour.
- **Coded raster is 1440 × 1088**, cropped to 1440 × 1080 by an 8-row conformance window; all four file sizes are exact multiples of 4096.

1.2 What the cameras actually are

CAM0 faces left and CAM5 right; both are wide fisheye with heavy vignetting and neither faces forward: the scene sweeps laterally across CAM5 at a median 18 px per frame at the 896×672 geometry raster (≈ 29 px at native 1,440; CAM0 23/37). That sets the difficulty: appearance depends on lateral lane position far more than on a forward camera, and a place is re-observed only when traversed in the same direction. A near/far scale test for lane offset flags 43.3%/43.5% of sampled pairs without persistence in time ($p=0.061$ on the opposite-sign test): no lane offset is established where the test can see.

Run B is 2.2× darker (mean luma 30.1/38.3 against 65.3/82.6), half as sharp, and has **rain on the lens that is static for the whole recording**: the droplets are crisp in both the temporal median and the temporal standard-deviation map, which only a fixed feature can be. That makes them measurable without a learned model, as $|\nabla(\text{temporal median})|/(\text{temporal std} + 1)$:

Run	Camera	Camera-attached	Never bright	Excluded
A	CAM0	5.34%	1.76%	7.15%
B	CAM0	4.37%	8.55%	12.99%
A	CAM5	1.23%	0.24%	1.50%
B	CAM5	9.92%	3.27%	13.43%

1.3 Route topology

Retrieval over the matcher's own descriptors finds a strong tail-to-head revisit in all four sequences (Run A A2604→180 at cosine 0.981 on CAM0, A2602→180 at 0.975 on CAM5; Run B B2534→134/0.938 and B2603→122/0.944): the route is a loop. It aliases internally too: of the 2,745,996 Run A frame pairs more than 150 frames apart, 8.5% exceed cosine 0.85 on CAM5 against 3.6% on CAM0, 2.4× the density of look-alike places (1.6–2.4× across cosine 0.80–0.90). And the A → B offset is not constant: dynamic time warping on an illumination-robust descriptor puts it between −112 and +12 frames, a **12.4-second drift** the annotator's independent labels reproduce (−89 to +13). A fixed offset is wrong by over ten seconds.

2 Method

2.1 Each stage answers an observed failure of the simpler one

Linear progress drifts 12.4 s; nearest-neighbour retrieval alone confuses the repeated awnings and foliage corridors, so route order and local verification are added; forced dense output cannot be right where nothing corresponds, so the method abstains.

Canonical decoded image → reliability mask → frozen DINOv2-S descriptor → full-route coarse similarity → monotonic path with a slope prior → full-cadence refinement → local RootSIFT/RANSAC → **accept or abstain**

The earlier sparse-anchor method (v1) placed 36 fixed Run A anchors per camera and interpolated only between adjacent accepted ones, so an isolated accept never propagated. The submitted method (v2) aligns every route frame through a globally optimised monotonic path, refines to full cadence, tests corridor uniqueness at four time scales, and verifies geometry with RootSIFT/RANSAC requiring 20 matches, 12 inliers, a 0.25 inlier ratio and support across a 4×3 grid. Both forbid departure–arrival wrap and interpolation across the seam, because the endpoints alias. The rebuild decodes from the HEVC with pinned colour rather than from 640×480 proxies, uses the measured mask, and runs geometry at a raster chosen by measurement on labelled-correct development pairs:

Geometry raster	CAM0 matches	CAM0 passed	CAM5 matches	CAM5 passed	CAM5 inlier ratio
448×336	22	2/6	12	1/6	0.71
640×480	31	4/6	21	1/6	0.62
896×672	48	6/6	40	5/6	0.46
1440×1080	85	4/6	86	2/6	0.33

Passing peaks at 896×672 and *falls again* at native, where fine foliage texture inflates the raw match count while the inlier ratio falls (CAM5 0.46→0.33, CAM0 0.61→0.39). Keypoint count alone would have chosen wrong.

2.2 Evaluation design

Eighty prediction-blind labels, 40 per camera, on Run A ordinals the 36 anchors never touched, stratified along the route with parked frames excluded. The page showed the query frame and a scrubbable Run B filmstrip and nothing else: no prediction, no score, a deliberately weak linear prior. The method was **frozen before the labels were imported** — every parameter hashed at 15:41:34 UTC, the seal written at 15:52:15 UTC citing it — and a verifier refuses to score if any live constant has drifted. That records *ordering*, not custody. v1 was frozen long before these labels existed.

3 Results, and two protocol defects

Method	Cam	Accepted	Coverage	±1	±2	±5	±10	Wilson 95% (±2)	strict ±0
v2 unified (submitted)	CAM0	32/32	100.0%	81.2%	84.4%	90.6%	96.9%	0.68–0.93	71.9%
v2 unified (submitted)	CAM5	32/37	86.5%	56.2%	68.8%	90.6%	96.9%	0.51–0.82	37.5%
v1 sparse-anchor	CAM0	17/32	53.1%	82.3%	88.2%	100.0%	100.0%	0.66–0.97	64.7%
v1 sparse-anchor	CAM5	4/37	10.8%	75.0%	75.0%	100.0%	100.0%	0.30–0.95	25.0%
v2 unified, targeted set	CAM0	21/21	100.0%	71.4%	81.0%	85.7%	100.0%	0.60–0.92	61.9%
v2 unified, targeted redo	CAM5	21/23	91.3%	38.1%	71.4%	85.7%	100.0%	0.50–0.86	14.3%

Precision at frame tolerances: the share of accepted match labels whose prediction lies within $\pm k$ frames of the annotator's acceptable interval (distance 0 inside it). One frame is 0.1 s, about a metre of travel, so ± 2 is roughly 2 m — an order of magnitude tighter than the 25 m or ± 10 -frames-at-1 FPS tolerances published place-recognition protocols use, which is why the ± 10 column is the one comparable in scale to them. Cameras are never pooled: they observe the same two traversals. Undetermined labels leave the denominator (§3.1); the strict ± 0 column is expanded in the last table.

3.1 A label-semantics correction, made after the fact

The page offered a match or “no usable match”, merging two facts: that Run B has no counterpart here, and that the annotator could not tell. All 11 such labels meant the second, checkably — queries were drawn inside the stretch both runs traversed (Run A 180–2616 into Run B 114–2621), so a genuine absence was impossible by construction. Undetermined queries therefore leave the denominator. The correction was made *after* the results were seen, applied identically to both methods, and raises the denser one more; the table's last column keeps the uncorrected reading.

3.2 A second blind set, drawn to test what the first one suggested

With the v2 labels spent, 48 fresh queries were sealed against a freeze naming the submitted mapping, the slope variant, the posterior and the slope gate, and scored once. Half target kinks and dark stretches, half are a uniform *corridor* control; targeting used the submitted mapping's own structure, so **only the corridor stratum is an unbiased route-wide rate**: $5/9 = 0.556$ on CAM0, against 0.619 (0.41–0.79) over all its match labels. **Its CAM5 half was withdrawn** for a defect found in the sealed labels themselves: the viewer displaced Run B by the measured pose offset instead of its negation, and the annotator's slider corrections correlate with the mapping's errors (TOOL_DEFECT_cam5.md). CAM0's offset is zero, so CAM0 stands. CAM5 was re-asked on 24 fresh queries under its own freeze, sign fixed, every row at the 40/12 px default: corridor $2/15 = 0.133$ (0.04–0.38), 0.143 (0.05–0.35) overall, 15/21 within two frames, and its errors are late-biased (median +2, 17:1 late:early).

Part of that bias is one pre-measured constant. The matcher's “same place” is image content; the labels' is a world position. The 18.5 px gap between the two — -1.79 frames at CAM5's sweep rate — was measured on the v2 labels before this set was drawn. Shifting every CAM5 prediction by the rounded -2 , the submitted file *unchanged*, gives $14/21 = 0.67$ (0.45–0.83) against

0.14 at 0 and 0.57 at -3 : overlapping intervals, a peak located rather than resolved. Two controls bound it. On the v2 CAM5 labels the same shift is **neutral** (0.375 at 0 and at -2), which is what a mixture predicts: those labels used the marker direction for some queries and the image layout for others (dx s.d. 29.8 px about a -56 px median). On CAM0, whose offset is zero, the sweep peaks at 0 (13/21 falling to 5/21). So: a convention constant on a consistently labelled set, neutral on the mixed one, and the two blind CAM5 readings (0.375, 0.133) are not distinguishable at $n=32/15$. A sensitivity reading on spent sets, not a held-out number; §4 attaches the constant.

The kink claim replicates on CAM0 in direction only (0.57 against 0.44) and *not* on CAM5 (0.80 against 0.87). The slope gate buys 0.619→0.714 on CAM0 for coverage 1.00→0.67 and costs precision on CAM5 (0.143→0.100); the exposure gate costs CAM0 precision (0.500), all 3/3 CAM0 dark labels being right. The unsubmitted slope variant beats the shipped mapping on both cameras (0.762/0.191).

3.3 Abstention: what the labels cannot say, and what can

The v2 set held no true no-correspondence case, so it could not evaluate abstention, and an earlier reading that abstention had collapsed is withdrawn. Each targeted set supplies one such query: all six methods accepted v3's, and 5 of 6 rejected the redo's. Withholding a stretch of Run B supplies more: frames whose partner lay inside it have none by construction. Detection at 60/150/400-frame holes runs 0.40/0.64/0.83 on CAM0, rising with the hole as it should, but *falls* on CAM5 (0.93/0.86/0.79) at 2.0% false alarms against 0.7%; 3 trials per cell, no interval: read the direction, not the digits. The route also supplies one *real* absence, found by looking at frames: an oncoming lorry fills the Run B CAM0 image at B240–270, so those descriptors describe a side panel, not a place. Both mappings drive straight through it — 30 accepted rows at confidence 1.00, at most 0.00021 no-correspondence probability from the posterior. Three code-level causes, verified by inspection: a monotonic path with pinned endpoints is always complete, so “no correspondence” is not representable; the sequence gate tests aliasing, not existence (its longest window passed 100% of frames); and geometry is propagated between sampled checks — reserving the top tier for frames actually checked collapses it from 2,366 to 156 on CAM0, so **93% of top-tier acceptances were never verified**.

3.4 Reading the two cameras

CAM0 is the stronger result: every determinable query answered, 84.4% within two frames and 0.719 strict, against 53.1% coverage for v1 — and of the fifteen queries only v2 answered, eleven are strictly right. CAM5 is weaker (v1 accepts 4 of 37), but its failure is a *localisation bias*, not place confusion: 68.8% within two frames and 90.6% within five, against 0.375 strict. Two measurements say why. **(i) The camera moved between the runs.** At the matcher's own pairs the CAM5 scene sits at $dx = -37.5$ px, $dy = -11$ px, at the annotator's pairs at $dx = -56$ px, constant along the route; CAM0 is 0/0 on both. A depth-profile test over ~ 190 pairs is **consistent with rotation** rather than travel (CAM5 -0.6 ± 1.5 frames from the depth-uniform position, CAM0 control -1.2 ± 0.7) but at SE 1.5 cannot separate 0 from -1.8 frames. **(ii) CAM5 is aimed low:** roughly 70% of every frame is near-field asphalt, and a label-free gradient-NCC sweep of the far field gives a median z against background of 3.4 on CAM5 against 6.8 on CAM0 — half the cross-run evidence everywhere.

3.5 A label-free signal, dense over the whole route

CAM0 and CAM5 face opposite sides, so their alignments come from disjoint scenes and fail near-independently. Comparing them is dense over 2,183 frames: median disagreement 1 frame, p95 11. Run *before any label existed*, it flagged A1068–1110, where five queries later proved undeterminable, inside a difficulty region covering 28% of the route (the A1050–1450 fence corridor, the A1500–1750 tree tunnel). It measures consistency, not correctness. It also judged the variants built once the labels were spent. Five input-side changes (greyscale, sky mask, both, parallax re-selection, a speed channel) move **nothing**: p95 stays at 11–12 frames. One change to the *pooling* does: an unsupervised VLAD over the same masked tokens (32 words learned from the route, nothing trained) takes p95 from 11 to 8 frames over 2,389 frames mapped by both cameras (2,183 before), lifts CAM5 coverage 0.816→0.898, reads 0.781 strict on the spent CAM0 labels against 0.719, and leaves the kinks (24/33) where they were: the mean pooling is a bottleneck, the input side is not, and the kinks belong to stage 3. No held-out number; not submitted.

3.6 Where the submitted mapping is least reliable

CAM5 throughout (12/32 strictly correct): usable for coarse retrieval, not frame-exact supervision. **Under-exposed frames:** the exposure flag fires on 304 of 2,622 Run B CAM0 frames (one stretch runs 177) and 136 of 2,615 on CAM5; 273 of 2,401 accepted CAM0 and 147 of 2,183 CAM5 rows have a flagged partner (130 strong plus 17 sequence-supported on CAM5). Dark matches dark almost regardless of place, so that error is *systematic*: 0/1 CAM5 blind dark labels are right against 3/3 on CAM0. **Implausible one-step jumps (“kinks”):** the first difference of the mapping is a local speed ratio, and the coarse search may not sustain more than $2\times$, so an increment of 6 in 0.1 s is unreachable — yet the mapping asserts 27 on CAM0 and 25 on CAM5, all accepted *strong*, against 1 per camera for the posterior (all four computed thresholds ship; at $K \geq 4$ it is 49/70 against 11/16). The cause is stage 3: greedy refinement behind a non-decreasing floor *with no slope prior*, so a collapsed structure score sends its argmax to the window edge and the floor keeps it there (coarse path step 3 against 11; 74 frames pinned at $+8$, 45 at -8). The *--refinement* slope variant (a Viterbi with a slope prior and a bounded step) has no kinks by construction; its case is cross-camera p95 11→9 at unchanged coverage, better structure scores in A880–940 (0.60 against 0.52) and both blind sets, at the cost of CAM0 closure 4/4→2/4. It stays unsubmitted: the submission was frozen before either blind set was opened; §4 gates the neighbourhood instead. **Anywhere abstention matters:** an empty row means none *under this method*, never an automatic negative.

4 Downstream policy for training or testing

4.1 Leakage-safe split, decided before anything else

Assign the split before sampling, descriptor extraction or calibration; group by route, traversal date, rig and *all* paired cameras, and keep the loop-equivalent departure and arrival in one group as distinct occurrences. Never randomise adjacent frames or treat CAM0 and CAM5 as independent. The present design prevents label tuning but remains transductive: one route, one pair of dates.

Action	Concrete rule for this data	Consumer protection
keep	Retain all 10,585 decoded images in the immutable source archive, addressed by display-order ordinal with decoder, colour matrix and source hash recorded.	Raw evidence and identity stay recoverable.
exclude	Exclude each run's parked prefix — Run A ordinals 0–174 plus transition, 180 frames per camera; Run B 114 (<code>route_phase_boundaries.csv</code>) — and exclude blind blocks from any tuning.	Stationary scenes cannot dominate, and labels cannot leak.
quarantine	Cluster near-duplicates only <i>after</i> split assignment; retain cluster size and weight $1/\text{size}$.	Adjacent frames cannot cross splits or dominate training.
quarantine	Automatic correspondences are weak positives: quarantine every segment with a blind false accept, and CAM5 for frame-exact use (it stays usable at ± 2 and as corroboration).	Model output cannot silently become ground truth.
quarantine	Demote out of the top tier any pair whose Run A or Run B frame carries <code>too_dark</code> or <code>occluded</code> (§3.6 counts them).	A dark-against-dark match is near-certain regardless of place.
quarantine	Slope gate: demote any top-tier row within 10 Run A frames of a one-step Run B increment ≥ 6 : 454 of 2,366 CAM0 and 320 of 1,881 CAM5 rows, shipped as <code>slope_gate</code> in the quality-flag table. Advisory on CAM5, where it cost precision.	A jump the mapping's own motion model forbids marks a neighbourhood it cannot justify.
exclude	Never convert an empty mapping to a hard negative. Only a reviewed no-match is a negative label.	Abstention is not confused with scene absence.
exclude	Blur plates and faces before any view leaves the archive. The stationary car-park segments, where residents' plates stay legible for ~ 23 s, are the highest PII density; the side cameras also record private frontages and pedestrians, including children.	Malaysia's PDPA 2010 applies, plus GDPR/UK GDPR for EU or UK processing.

4.2 The per-frame manifest: a schema, and what ships against it

`frame_manifest_schema.json` is a closed schema, not a shipped table; what ships against it is `frame_quality_flags.csv` plus the mapping CSVs' own columns. Its fields: **identity** (policy version, run, camera, source SHA-256, decoder, colour matrix and range, coded and display raster, display-order ordinal); **time** (run start/end, and per-image capture time *explicitly null* with its reason: both cameras of a run share one byte-identical sidecar that cannot describe both — Run B decodes 2,622 and 2,615 frames against one 2,663-row log); **quality** (route phase, mask version, `too_dark` with its two luma measurements, `occluded`, `slope_gate`, temporal cluster, sample weight); **governance** (route/traversal group, split, boundary buffer, paired camera); **correspondence** (partner or null, acceptable interval, typed status, label source, freeze hash, known failures, and the camera-to-world convention offset: CAM0 0, CAM5 -2 frames for world-position use — measured on v2, validated on v3b, neutral on v2 itself). It forbids four silent assumptions: sidecar row i as decoded image i , cameras synchronised by ordinal, coverage as accuracy, confidence as probability. Two facts a consumer must know: confidence is an ordinal rank whose 0.50 tier never occurs, and the mapping is **many-to-one** (34% of CAM0's mapped rows share a partner, max 12; 52%, max 15, on CAM5) — expected where Run B is slower, fatal to a consumer keying on `runB_frame`.

4.3 A posterior formulation, built but not claimed

The three structural gaps in §3.3 share one cause: the method had no posterior. A hidden Markov model over Run B position with an explicit *null* state tests that reading (emissions are likelihood ratios against each query's own similarity row; the §3.3 holdout runs against it at a threshold set on synthetic sequences, never on labels). Loop closure (one place in Run A must map to one in Run B) reaches 2/5 consistent revisit pairs on CAM0 and 1/5 on CAM5 against 4/4 and 0/3 for the submitted method, unchanged by coupling; the denominators differ because the submitted mapping abstains on some revisit ends. Its suspension-shake channel needed correcting first: both cameras ride one chassis, so a jolt must fire in both streams at once, and that label-free test *selects among three constructions*: raw frames 0 of 28 on Run B, temporal-mean division 4 of 33 ($3.85\times$ chance, 9 of 45 on Run A), median subtraction 0 of 8. The 7 against 22 event asymmetry was therefore an artefact (16 against 24 corrected), the ratio 2.83 is withdrawn, and 97.8/95.1% of the mapping is unchanged either way.

Both freezes named it, so it has a held-out score: 0.714 on CAM0 and 0.174 on CAM5 against 0.619 and 0.143 submitted; calibration error 0.102 and 0.295 (the second overconfident); p95 disagreement 11 frames to 4. Not submitted: the submission was frozen first, and its null state accepted both sets' `no_correspondence` query, alone among six on the redo's.

4.4 The strict single-frame reading (v2 labels, tolerance zero)

Method	Cam	Accepted	Correct	Precision	Wilson 95%	Coverage	Median err	Within 2	As-labelled
v2 unified (submitted)	CAM0	32/32	23	0.719	0.55–0.84	100.0%	1.0	26/32	0.575
v2 unified (submitted)	CAM5	32/37	12	0.375	0.23–0.55	86.5%	2.0	21/32	0.343
v1 sparse-anchor	CAM0	17/32	11	0.647	0.41–0.83	53.1%	1.0	14/17	0.647
v1 sparse-anchor	CAM5	4/37	1	0.250	0.05–0.70	10.8%	1.5	3/4	0.250

A hit only inside the annotator's interval (width 0 to ± 1); *As-labelled* keeps the uncorrected denominator of §3.1. **Submitted:** this PDF, the mapping CSVs, `assessment.ipynb`, scripts and tests, three sealed blind sets with seals and freezes, `FAILURE_ANALYSIS.md`, `README.md`, `DATA_POLICY.md`, the quality-flag table. Raw video, weights and caches excluded.